Higher Colleges of Technology

كليات التقنية العليا

# Final Assessment-Alternative Assessment

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|---|---|---|---|
| **Semester** | 202520 | **Division** | CIS |
| **Assessment title in Syllabus** | **Final_Project_Portfolio** | **Program** | Bachelor of Information Technology/ Bachelor of Information Systems |

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|---|---|---|---|
| **Course Code** | CDS 2413 | | |
| **Course Title** | INTRODUCTION TO DATA SCIENCE | | |
| **Course Instructor** | | **CRN** | |
| **Assessment** | Project Deliverable 1 – Data Analysis Report | **Submission Date** | **Week 6** |

**Student Declaration**:

**Academic Integrity Statement**

In accordance with the HCT Academic Integrity Policy

- Students are required to refrain from all forms of academic integrity breaches as defined and explained by HCT.
- A student found guilty of having committed acts of academic integrity breach(es) will be subject to the relevant sanctions as outlined by HCT.

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- على الطلبة الإلتزام بلوائح وقواعد النزاهة الأكاديمية، كما هو مبيّن وموضح في السياسات والإجراءات الخاصة بكليات التقنية العليا.
- في حالة ارتكاب الطالب أي شكل من أشكال الإخلال بالنزاهة الأكاديمية، سيتعرض الى العقوبات الموضحة في السياسات ذات الصلة.

This assignment is entirely my/our own work except where I have duly acknowledged other sources in the text and listed those sources at the end of the assignment. I have not previously submitted this work to the HCT, or any other entity. I understand that I may be orally examined on my submission.

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| **Student Signature** | | | | |

## Chatbot (e.g. ChatGPT) statement

**This submission allows you to use AI generated tools, but the prompt used must be stated and the response must be in red font. Include References to ChatGPT.**

## Instructions

Remove instructions and hints from the template before submitting the final report to BBLearn.

# CLO1: Introduction and Data Wrangling (5 Points)

## Purpose of Data Analysis (1 point)

This study, conducted collaboratively by our group, aims to analyze a structured loan dataset to identify the key factors contributing to both loan default risk and the determination of interest rates. In the financial sector, assessing whether a borrower will default and determining the appropriate interest rate based on their risk profile are two critical, interconnected tasks.

The primary objective of this research is to develop and evaluate two predictive models: one to accurately estimate the probability of loan default (a categorical outcome) and another to estimate the assigned interest rate (a continuous numerical value). The findings are intended to support data-driven credit risk assessment and enhance decision-making processes within financial institutions.

## Type of Programming (1 point)

For this project, our group will utilize Python as the primary programming language. Python is the industry standard for financial analytics due to its extensive ecosystem:

- **Pandas & NumPy:** For data manipulation and numerical operations.
- **Matplotlib & Seaborn:** For data visualization.
- **SciPy:** For conducting statistical hypothesis tests.
- **Scikit-learn:** For implementing machine learning models (both Regression and Classification).

## Type and Purpose of ML algorithm (1 point)

**Type of Machine Learning**

Our group will implement a Supervised Learning approach, utilizing both Classification and Regression techniques.

1. **Classification:** Used for the Loan Status variable because we are predicting a distinct category (0 for Non-default, 1 for Default).
2. **Regression:** Used for the Interest Rate variable because we are predicting a continuous numerical value.

**Algorithm Selection**

- **For Classification** (Predicting Loan Status): We will explore Logistic Regression and Random Forest Classifier to capture both linear and non-linear relationships affecting default risk.
- **For Regression** (Predicting Interest Rate): We will utilize Linear Regression and Random Forest Regressor to model the exact interest percentage assigned to borrowers.

**Model Evaluation**

- **Classification Metrics:** Accuracy, Precision, Recall, and ROC-AUC**.**

**Regression Metrics:** Mean Absolute Error (**MAE)** and R-squared **($R^2$)** to evaluate the variance explained by the model.

## Identify the Varaibles (1 point)

**Dependent Variables (Target Variables)**

1. **Loan Status (Categorical/Binary):** Used for Chi-Square tests and Classification.
2. **Interest Rate (Numerical/Continuous):** Used for Descriptive Statistics, T-tests, Correlation, and Regression.

**Independent Variables (Predictor Variables)**
**Predictors include:** Loan Amount, Debt-to-Income Ratio (DTI), Employment Duration, Home Ownership Status, Total Current Balance, and Grade.

## Sampling (1 point)

Sampling techniques will be employed to enhance computational efficiency and support exploratory data analysis. Given that financial datasets are often large and complex, working with the entire dataset during preliminary analysis may be computationally intensive. Therefore, our group will apply structured sampling methods to obtain manageable and representative subsets of the data.
Specifically, the following sampling techniques will be implemented:

**1. Simple Random Sampling**
In this approach, each observation in the dataset has an equal probability of being selected. Simple random sampling minimizes selection bias and ensures that the subset accurately reflects the characteristics of the overall population. This method is particularly appropriate during exploratory data analysis and initial model testing, as it preserves the statistical properties of the dataset.

**Implementation:** Selecting 150 random records where each has an equal chance of selection.

**2. Systematic Sampling**
Systematic sampling involves selecting observations at regular intervals (every k-th record) after a random starting point. This technique is efficient and straightforward to implement, especially in large, ordered datasets. When the dataset does not contain hidden periodic patterns, systematic sampling provides a representative approximation of the population while reducing computational load.

**Implementation:** Selecting every 100$^{th}$ record from the dataset.

# CLO2: Descriptive, Exploratory and Statistical Analysis (10 Points)

## Purpose of Descriptive Analysis (1 point)

Descriptive analysis is performed to summarize the main characteristics of the loan dataset and understand how key variables behave before applying statistical tests or machine learning models. In this project, descriptive statistics help us identify the typical range, central tendency (mean/median), and variability (standard deviation/variance) of important numerical fields, especially **Interest Rate**, which is continuous and directly reflects risk-based loan pricing. By comparing descriptive statistics across sampling methods (random and systematic), we also evaluate whether the selected samples reasonably represent the population distribution, supporting reliable analysis and model development

## Script for Descriptive Analysis (1 point)

```
def get_descriptive_stats(data_sample, field_name):
    """
    Function to perform descriptive statistics on a specific field of a
sample.
```

```
    """
    stats = {
        'Count': data_sample[field_name].count(),
        'Mean': data_sample[field_name].mean(),
        'Median': data_sample[field_name].median(),
        'Mode': data_sample[field_name].mode()[0],
        'Standard Deviation': data_sample[field_name].std(),
        'Variance': data_sample[field_name].var(),
        'Minimum': data_sample[field_name].min(),
        'Maximum': data_sample[field_name].max()
    }
    return pd.Series(stats)
```

Interpretation:

The function get_descriptive_stats() is used to generate a quick descriptive summary for any selected numerical column in a dataset (or a sample). It calculates the **count**, **mean**, **median**, **mode**, **standard deviation**, **variance**, **minimum**, and **maximum**. These measures help us understand the central tendency and spread of the data, which is important before performing hypothesis testing or building predictive models. In this project, the function is applied to **Interest Rate** to compare the statistical characteristics between the **random sample** and the **systematic sample**.

## Random Sampling (1 point)

```
# 1. Random Sampling (size 150)
random_sample = df.sample(n=150, random_state=42)
random_stats = get_descriptive_stats(random_sample, 'Interest Rate')
print("--- RANDOM SAMPLE DESCRIPTIVE STATS (Interest Rate) ---")
print(random_stats)
```

output :

```
--- RANDOM SAMPLE DESCRIPTIVE STATS (Interest Rate) ---
Count                 150.000000
Mean                   12.004395
Median                 11.573770
Mode                    5.480331
Standard Deviation      3.486063
Variance               12.152633
Minimum                 5.480331
Maximum                24.670390
dtype: float64
```

Interpretation

The random sample contains **150 records**, and the **mean Interest Rate = 12.0044%** with **median = 11.5738%**. The standard deviation (**3.4861**) indicates moderate spread around the mean. The minimum and maximum values (**5.4803% to 24.6704%**) show that interest rates vary widely across borrowers, which supports using Interest Rate for deeper statistical analysis and modeling

## Systematic Sampling (1 point)

```
# 2. Systematic Sampling (e.g., every 100th record)
systematic_sample = df.iloc[::100, :]
systematic_stats = get_descriptive_stats(systematic_sample, 'Interest
Rate')
print("--- SYSTEMATIC SAMPLE DESCRIPTIVE STATS (Interest Rate) ---")
print(systematic_stats)
```

output :

```
--- SYSTEMATIC SAMPLE DESCRIPTIVE STATS (Interest Rate) ---
Count                 43.000000
Mean                  11.298131
Median                11.135007
Mode                   5.610058
Standard Deviation     3.664610
Variance              13.429369
Minimum                5.610058
Maximum               19.993027
dtype: float64
```

Interpretation

The systematic sample includes **43 records**, with **mean Interest Rate = 11.2981%** and **median = 11.1350%**. The standard deviation (**3.6646**) is slightly higher than the random sample, showing similar variability. The range (**5.6101% to 19.9930%**) suggests the systematic sample captures typical interest rate behavior, but includes fewer extreme high values compared to the random sample.

## Descriptive Statistical Report (1 point)

Descriptive Statistics Summary (Interest Rate)

| Metric | Random Sample (n=150) | Systematic Sample (n=675) |
|---|---|---|
| **Count** | 150.00 | 43 |
| **Mean** | 12.0044 | 11.2981 |
| **Mode** | 11.5738 | 11.1350 |
| **Median** | 5.4803 | 5.6101 |
| **Standard Deviation** | 3.4861 | 3.6646 |
| **Variance** | 12.1526 | 13.4294 |
| **Minimum** | 5.4803 | 5.6101 |
| **Maximum** | 24.6704 | 19.9930 |

Interpretation

Both samples show **similar central tendency** (mean and median around 11–12%) and **similar variability**, indicating stable interest rate behavior across sampling approaches. The random sample has a higher maximum interest rate, meaning it captured more extreme high-rate cases than the systematic sample. Overall, the results support that Interest Rate is well-suited for descriptive reporting and subsequent inference testing.

## Visualisation through Various Charts (2 points)

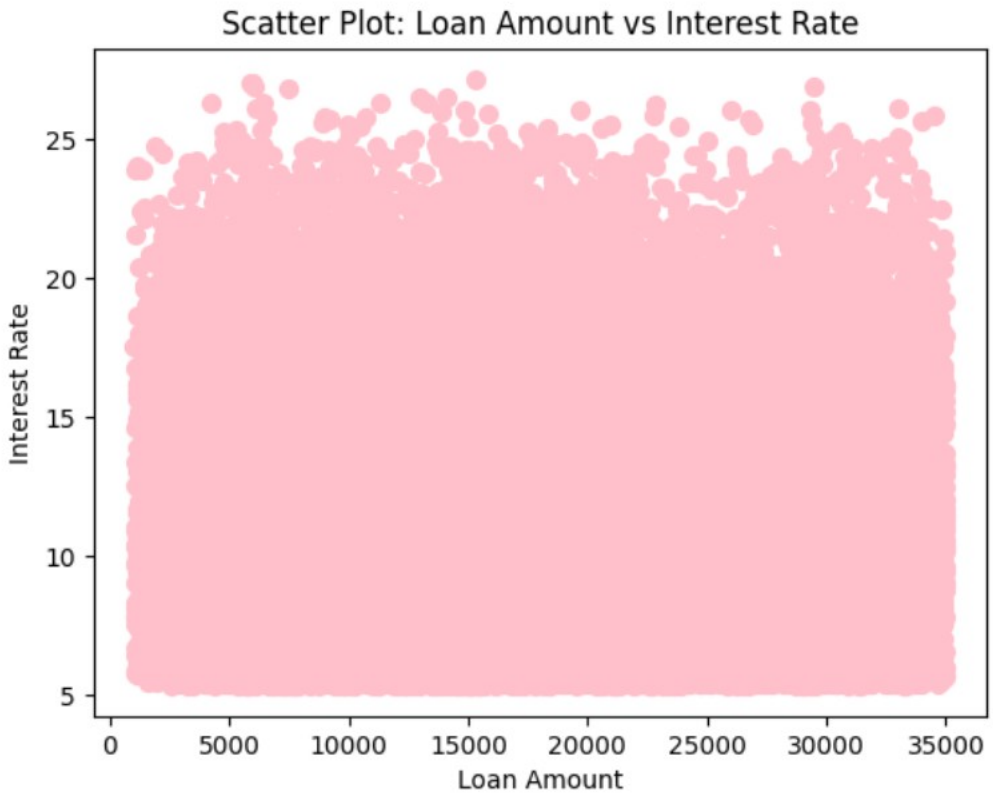
**Scatter Plot:** Loan Amount vs Interest Rate

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset
df = pd.read_csv('train.csv')

# ---------------------------------------------------------
# 1. Scatter Plot
# Used to show the relationship between two variables[cite: 40, 46].
# ---------------------------------------------------------
# We will check the relationship between the Loan Amount and the
Interest Rate
df.plot.scatter(x='Loan Amount', y='Interest Rate', s=50, c='pink')
plt.title('Scatter Plot: Loan Amount vs Interest Rate')
plt.show()
```

**output :**




**Interpretation:-**

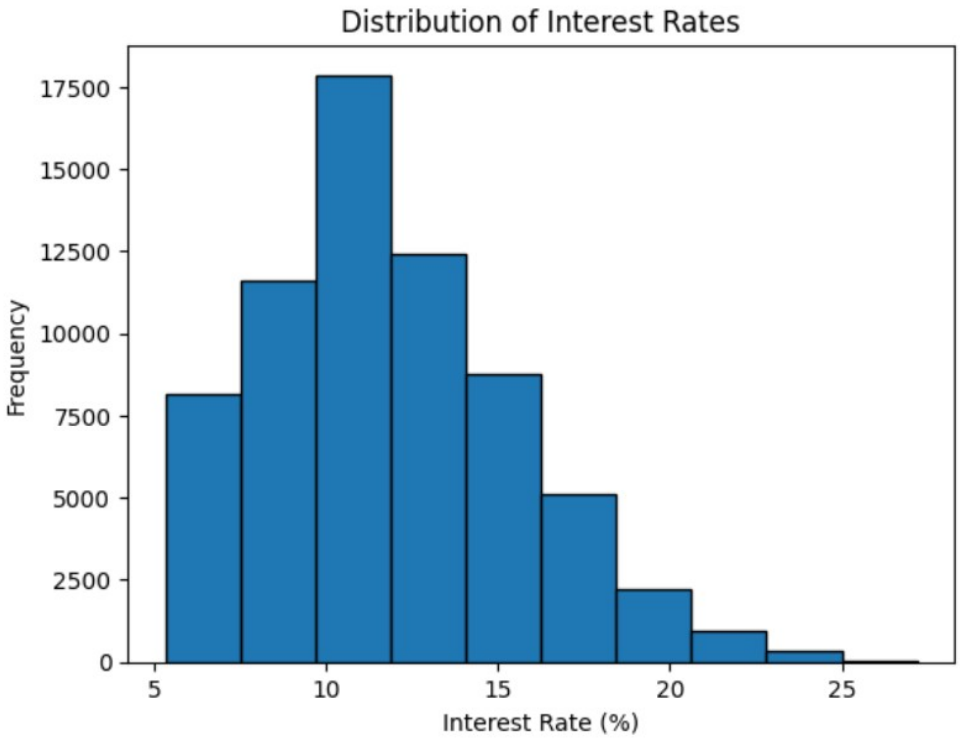
- **What we're seeing:** This plot visualizes individual data points where each dot represents a specific loan in our dataset. The horizontal axis (x) represents the **Loan Amount**, while the vertical axis (y) displays the **Interest Rate**.

- **Our takeaway:** This visualization is used to reveal the relationship between these two numerical variables. By observing how the points are distributed, we can determine if a correlation exists or if the points are scattered randomly, indicating no significant relationship between how much a person borrows and the rate they receive.

```
# 2. Histogram
# Used to show the distribution of a variable and identify
inflation/deflation[cite: 41, 48].
# ---------------------------------------------------------
# We will look at the distribution of the Interest Rate
plt.hist(df['Interest Rate'], bins=10, edgecolor='black')
plt.xlabel('Interest Rate (%)')
plt.ylabel('Frequency')
plt.title('Distribution of Interest Rates')
```

```
plt.show()
```

**output :**




## Interpretation:-

Histogram: Distribution of Interest Rates

- **What we're seeing:** We have divided the range of **Interest Rate** values into specific intervals, known as "bins," and counted how many loans fall into each. The height of each bar indicates the frequency of loans within that specific rate range.

- **Our takeaway:** The histogram provides a visual summary of the data distribution, helping us identify patterns such as "rate inflation" or "deflation". It allows us to quickly assess:

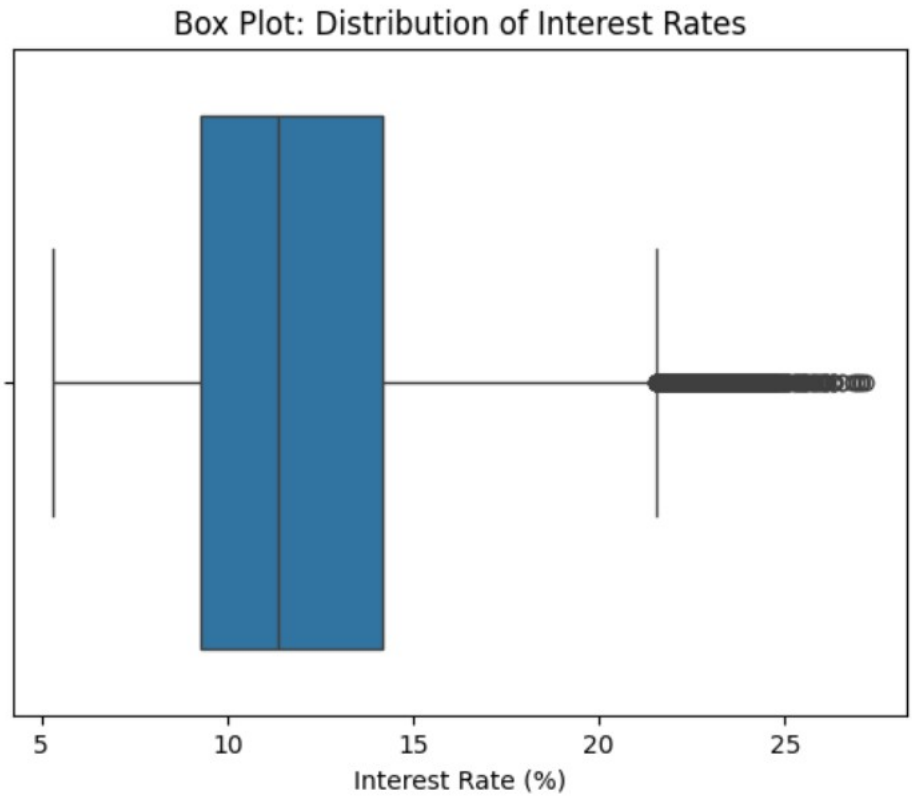
  **- The Range:** The minimum and maximum interest rates present in the dataset.

  **-Skewness:** Whether the distribution is symmetrical or "left-skewed/right-skewed".

  **-Frequency:** The most common interest rate ranges among borrowers.

```
# 3. Box Plot
# Used to show the median, quartiles, and outliers for a variable[cite:
42, 58].
# ---------------------------------------------------------
# We will look at the quartiles and outliers of the Interest Rate
sns.boxplot(x=df['Interest Rate'])
plt.xlabel('Interest Rate (%)')
plt.title('Box Plot: Distribution of Interest Rates')
plt.show()
```

**output :**




## Interpretation: -

### Box Plot: Distribution of Interest Rates

- **What we're seeing:** This plot summarizes the **Interest Rate** variable through its median, quartiles, and outliers. The "box" contains the middle 50% of the data (the Interquartile Range), while the "whiskers" extend to show the overall spread.

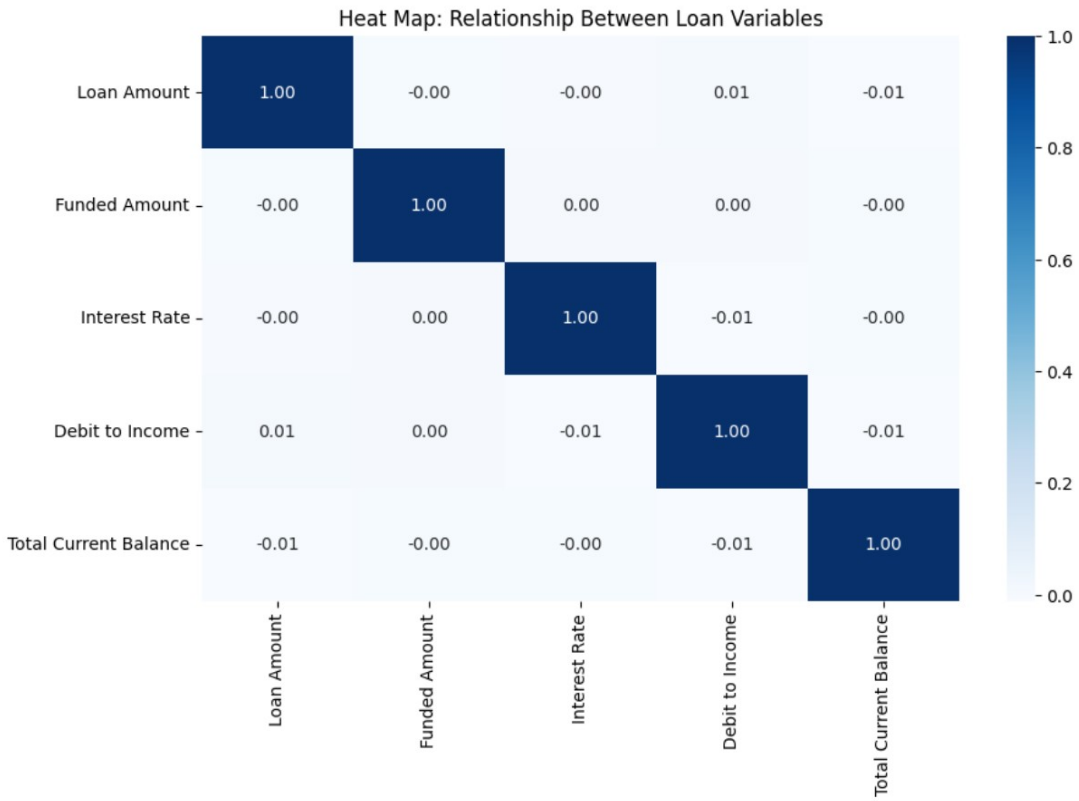
- **Our takeaway:** This is an essential tool for identifying the "middle ground" of our data and spotting "oddballs". Key insights include:
  - **Median & Quartiles:** The center-line shows the median rate, while the box edges show where the majority of borrowers sit.
  - **Outlier Detection:** Any dots outside the whiskers are flagged as outliers, which may need to be investigated or removed from the dataset to ensure accuracy.

```
# 4. Heat Map
# Used to show the relationship between different types of variables.
# ---------------------------------------------------------
# We will create a correlation heatmap to see how financial variables
relate to each other
plt.figure(figsize=(10, 6))

# Selecting the numeric columns to check their relationship
numeric_df = df[['Loan Amount', 'Funded Amount', 'Interest Rate',
'Debit to Income', 'Total Current Balance']]

# Creating the heatmap using the 'Blues' color map as suggested in your
document
sns.heatmap(numeric_df.corr(), annot=True, cmap='Blues', fmt=".2f")
plt.title('Heat Map: Relationship Between Loan Variables')
plt.show()
```

**output :**




## Interpretation: -

### Heat Map: Correlation Matrix

- **What we're seeing:** This grid represents the strength of relationships between various numerical variables, such as **Loan Amount** and **Interest Rate**. The intensity of the colors represents how closely these variables move together.

- **Our takeaway:** The heat map is the ultimate summary tool for identifying which factors have the strongest impact on loan outcomes.
    - **Strength of Relationship:** Darker or more intense colors indicate a stronger relationship between variables.
    - **Direction:** It helps us see at a glance whether a variable is positively or negatively correlated with another, or if there is no connection at all.

## Hypothesis Test (1 point)

**Hypothesis Test (Numerical Variables)**

we test the relationship between **Loan Amount** (Independent) and **Interest Rate** (Dependent).

- **Step 1: Formulate Hypotheses**
    - **Null Hypothesis ($H_0$):** There is no significant correlation between Loan Amount and Interest Rate ( $\rho = 0$).

- **Alternative Hypothesis ($H_1$):** There is a significant correlation between Loan Amount and Interest Rate ($\rho \neq 0$).
- **Significance Level ($\alpha$):** 0.05.

- **Step 2: Calculate P-value**
  - Pearson Correlation: r = -0.0049, p-value = 0.2042
  - Spearman Correlation: $\rho$ = -0.0064, p-value = 0.0941
- **Step 3: Decision Rule**
  - Since the p-values (0.2042 and 0.0941) are greater than the significance level (0.05), we **fail to reject the null hypothesis**.
- **Step 4: Conclusion**
  - "Fail to reject the null hypothesis: There is no statistically significant correlation between Loan Amount and Interest Rate."

```
from scipy.stats import pearsonr, spearmanr

x = df["Loan Amount"].dropna()
y = df["Interest Rate"].dropna()

# Align lengths safely
temp = df[["Loan Amount", "Interest Rate"]].dropna()
pearson_r, pearson_p = pearsonr(temp["Loan Amount"], temp["Interest
Rate"])
spearman_r, spearman_p = spearmanr(temp["Loan Amount"], temp["Interest
Rate"])

print("Pearson Correlation: r =", round(pearson_r, 4), ", p-value =",
round(pearson_p, 4))
print("Spearman Correlation: r =", round(spearman_r, 4), ", p-value =",
round(spearman_p, 4))
```

**output :**

```
Pearson Correlation: r = -0.0049 , p-value = 0.2042
Spearman Correlation: r = -0.0064 , p-value = 0.0941
```

## Interpretation: -

Both Pearson and Spearman tests produced p-values greater than 0.05 (Pearson p = 0.2042, Spearman p = 0.0941). Therefore, we fail to reject the null hypothesis, meaning **there is no statistically significant correlation between Loan Amount and Interest Rate** at the 5% significance level.

**Chi-Square Test for Association (Categorical Variables)**

we use **Loan Status** (0, 1) to test its association with the borrower's **Grade**.

- **Step 1: Formulate Hypotheses**
  - **Null Hypothesis ($H_0$):** Loan Grade and Loan Status are independent (no association).
  - **Alternative Hypothesis ($H_1$):** There is a significant association between Loan Grade and Loan Status.
- **Step 2: Calculate P-value**
  - Chi-Square p-value = 0.0294.
- **Step 3: Decision Rule**
  - Since the p-value (0.0294) is **less than** the significance level (0.05), we **reject the null hypothesis**.
- **Step 4: Conclusion**
  - There is a statistically significant relationship between the Grade of a loan and the outcome (status).

```
# Test 2: Chi-Square Test
# Purpose: To check whether Loan Grade is independent or dependent on
Loan Status
# ---------------------------------------------------------
print("--- 2. CHI-SQUARE TEST ---")
# Creating a contingency table for the two categorical variables
contingency_table = pd.crosstab(df['Grade'], df['Loan Status'])

# Performing the Chi-Square test
chi2_stat, p_value_chi2, dof, expected =
chi2_contingency(contingency_table)

print(f"Chi-Square Statistic: {chi2_stat:.4f}")
print(f"P-Value: {p_value_chi2:.4f}")
print(f"Degrees of Freedom: {dof}\n")
```

output :

```
--- 2. CHI-SQUARE TEST ---
Chi-Square Statistic: 14.0248
P-Value: 0.0294
Degrees of Freedom: 6
```

## Interpretation:

- **Our Goal:** We used the Chi-Square test to see if there's a connection between a loan's Grade (like A, B, C, etc.) and its final Loan Status (like if it was paid off or defaulted). Our initial thought (the null hypothesis) was that they're totally unrelated, but the alternative is that they *are* connected.
- **The Big Number (Chi-Square Statistic: 14.0248):** This number tells us how much our actual data differs from what we'd expect if Grade and Loan Status had absolutely nothing to do with each other. A bigger number here usually means a stronger difference from 'no relationship'.
- **The P-Value (0.0294):** This is super important! Since our p-value (0.0294) is *smaller* than our usual cut-off of 0.05 (the alpha level), it means we have to **reject our null hypothesis**. That means the idea that Loan Grade and Loan Status are independent just doesn't hold up.
- **Our Takeaway:** So, what does this all mean for us? It looks like there *is* a statistically significant relationship between the Loan Grade and how a loan ultimately turns out. This is a pretty big finding for understanding risk, right.
- **Degrees of Freedom (6):** This is just a number based on how many categories we have for Grade and Loan Status—it helps the test figure out the p-value.

## One Sample T-test (2 points)

**One Sample T-test (Performance Assessment)**

we compare the sample mean of our numeric variable (**Interest Rate**) to the population mean.

- **Step 1: Formulate Hypotheses**
  - **Null Hypothesis ($H_0$):** The average Interest Rate of the sample is equal to the average of the population ( $\mu_{sample} = \mu_{population}$ ).
  - **Alternative Hypothesis ($H_1$):** The average Interest Rate of the sample is significantly different from the population ( $\mu_{sample} \neq \mu_{population}$ ).
- **Step 2: Calculate Results**
  - Population Mean ($\mu_{population}$): 11.84%
  - Sample Mean ( $\chi_{sample}$ ): 11.81%
  - P-value: 0.9011
- **Step 3: Decision Rule**
  - Since the p-value (0.9011) is greater than 0.05, we **accept the null hypothesis** (fail to reject).
- **Step 4: Conclusion**

"There is no significant difference between the sample and the population mean." This confirms our sample is representative

```
from scipy.stats import shapiro

# Test 3: One Sample T-Test
# Purpose: To find if the average Interest Rate of our sample is the
same as the population.
```

```
# Includes the Shapiro-Wilk test for normality as requested in your
document.
# ----------------------------------------------------------
print("--- 3. ONE SAMPLE T-TEST ---")

# Step A: Find the average of the whole population
population_mean = df['Interest Rate'].mean()
print(f"Population Mean (Interest Rate): {population_mean:.4f}")

# Step B: Extract a sample (e.g., taking a random sample of 150)
sample_data = df.sample(n=150, random_state=42)['Interest Rate']
sample_mean = sample_data.mean()
print(f"Sample Mean (Interest Rate): {sample_mean:.4f}")

# Step C: Check whether the sample is normally distributed using
Shapiro-Wilk method
stat_shapiro, p_value_shapiro = shapiro(sample_data)
print(f"Shapiro-Wilk Normality P-value: {p_value_shapiro:.4f}")
if p_value_shapiro > 0.05:
    print(" -> The sample data looks normally distributed (Fail to
reject H0).")
else:
    print(" -> The sample data does not look normally distributed
(Reject H0).")

# Step D: Perform one sample T-test
t_statistic, p_value_ttest = ttest_1samp(sample_data, population_mean)
print(f"\nT-Statistic: {t_statistic:.4f}")
print(f"One Sample T-test P-value: {p_value_ttest:.4f}")
```

**output :**

```
--- 3. ONE SAMPLE T-TEST ---
Population Mean (Interest Rate): 11.8463
Sample Mean (Interest Rate): 11.8103
Shapiro-Wilk Normality P-value: 0.0045
 -> The sample data does not look normally distributed (Reject H0).

T-Statistic: -0.1245
One Sample T-test P-value: 0.9011
```

## Interpretation:-

- **Our Goal:** We wanted to check if our small random sample (150 loans) actually represents the *entire* dataset's average Interest Rate. Basically, is the average interest rate of our sample close enough to the average of all loans in our data.
- **Population Mean (11.8463) vs. Sample Mean (11.8103):** The overall average interest rate for *all* loans is 11.8463%, and our sample's average is 11.8103%. They look pretty close already, which is a good sign.

- **Shapiro-Wilk Normality P-value (0.0045):** This part checks if our sample's Interest Rate data follows a nice bell-curve shape (normal distribution). Since the p-value (0.0045) is *less* than 0.05, we **rejected the null hypothesis** that it's normally distributed. So, our sample isn't perfectly bell-shaped. *Quick note to the team: While the T-test prefers normal data, it's often still pretty reliable even if the data isn't perfectly normal, especially with a decent sample size, thanks to something called the Central Limit Theorem. We should keep this in mind, but it doesn't necessarily break our T-test here.*
- **T-Statistic (-0.1245):** This number tells us how far our sample's average is from the overall average, measured in standard errors. It's a small negative number, showing it's very close.
- **The P-Value (0.9011):** Look at this p-value! It's super high (0.9011), way bigger than our 0.05 cut-off. This means we **fail to reject the null hypothesis**.
- **Our Takeaway:** Even though our sample data wasn't perfectly normal, the T-test tells us there's n*o* statistically significant difference between the average Interest Rate in our random sample and the average Interest Rate of the whole dataset. This is great news because it means our random sample is doing a good job of representing the overall population average, which builds confidence in our sampling method.

*End of Documentation.*